Senior Embedded Software Leader

Norberto Muñoz Gómez

Senior embedded software leader with 20+ years of experience in software development, control systems, and mission-critical engineering across automotive, telecom, and industrial domains. Combines hands-on software engineering expertise in C/C++, agile development, verification, DevOps-oriented delivery, and complex system integration with strategic leadership in software team management, coaching, resource planning, and delivery ownership. Has led software teams and technical initiatives in environments requiring high reliability, structured development processes, and close collaboration with technical, internal, and external stakeholders. Fluent in Swedish and English, with a PhD-level background in automation and robotics and a leadership style focused on collaboration, continuous learning, and sustainable engineering performance.

AVAILABILITY According to agreement · norberto.munoz@mpyascitech.com · +46 73 321 3394

Tech competence

MAIN SKILLS

- Embedded Software Engineering
- V-model development
- Complex system integration
- Agile development
- DevOps
- Mission-critical software
- Control Systems
- AI-assisted development
- Technical innovation leadership

OTHER SKILLS

- SW teams management
- Coaching and mentoring
- Career plan support
- Cross-functional multicultural collaboration
- Continuous learning
- Budget and resources negotiation and planning
- Strategic growth support

TOOLS

- C / C++ / C#
- Matlab/Simulink
- Python
- AUTOSAR
- Git
- Gerrit/BitBucket
- Jira
- GH Copilot
- ChatGPT/Claude

Experience

Volvo Group – Autonomous Functions (via Mpya Sci & Tech)
Oct 2021 – Sep 2024

Innovation Leader & Senior Software Function Developer

- Led AI tooling adoption across three teams, including rollout, training, coaching, environment definition, and license onboarding.
- Coordinated cross-team initiatives to apply AI-enabled automation to SIL and HIL testing, supporting innovation in embedded software verification.
- Worked with licensing and engineering stakeholders to enable compliant daily use of AI tools in safety-critical embedded software development.
- Developed fail-operational embedded C and AUTOSAR control software for autonomous-enabled trucks within a structured V-model process.
- Collaborated with test engineers and system architects to ensure robust, safe, and ISO 26262-compliant software delivery

Tools & Skills: C, C#, Python, AUTOSAR, ISO 26262, V-model, CI/CD, Jira, GH Copilot, VolvoGPT

Ericsson (via Mpya Sci & Tech)
Oct 2021 – Aug 2024

Software Engineer, C++ Automation and System Configuration

- Developed C++ software solutions to automate setup and configuration of radio base stations with multiple units and radio access technologies.
- Worked in an agile DevOps-oriented environment using Git, Gerrit, Jira, Jenkins, GoogleTest, VS Code, and Eclipse.
- Contributed to automation and configuration of complex system-level software environments requiring structured troubleshooting and cross-functional collaboration.
- Supported software delivery practices aligned with agile development, continuous integration, and reliable configuration of telecom system environments.

Tools & Skills: C++, GoogleTest, Jenkins, Git, Gerrit, Jira, Agile, DevOps, VS Code

General Electric
Sep 2005 – Jan 2020

Technical Software Lead
Jan 2014 – Jan 2020

- Led technical software work for embedded control systems used in turbine engine applications across marine and industrial domains.
- Supported technical direction and software engineering decisions in control applications requiring reliability, structured development, and robust system behavior.
- Developed tools to migrate control software applications from UX-beacon7 to Windows-Simulink, supporting software engineering modernization.
- Designed and implemented control logic improvements for turbine engine systems, including coke formation purge and fleet emissions auto-tuning.
- Applied V-model, 6Sigma DMAIC, and TOPS 8D practices to structure problem solving, defect investigation, and reliable software delivery

Tools & Skills: Matlab/Simulink, C, Toolbox, V-model, 6sigma DMAIC, TOPS 8D, Unix

Team Manager, Software Development
Jan 2011 – Jan 2014

- Led a software development team, grew the team through hiring, managed resources, defined team goals, and ensured delivery against established targets.
- Supported career plans, carried out performance appraisals, resolved conflicts, and strengthened team execution.
- Balanced resource planning, team performance, and stakeholder expectations in a mission-critical engineering context.
- Provided people leadership in an embedded control software environment requiring reliable delivery and structured collaboration.
- Developed leadership practices around team goals, individual development, communication, and delivery follow-up in a technical engineering organization

Tools & Skills: GE guidelines for interviewing, hiring, leadership skills, career growth and performance appraisal

Embedded Control Software Developer
Sep 2005 – Jan 2011

- Developed embedded control software for turbine engines used in marine and industrial applications.
- Applied formal requirements, functional validation, unit testing, software simulator testing, HIL validation, performance analysis, robustness analysis, and stability analysis.
- Supported software delivery from budget and resource negotiation through validation in test cells and customer-site environments.
- Worked across development, verification, validation, and customer-facing environments, building a strong foundation in reliable embedded control software delivery.

Tools & Skills: Fortran, Perl, Kornshell, Unix, V-model, Unit Test, SIL, HIL

Education & Languages

DEGREE

PhD and DEA in Advanced Automation and Robotics
Universidad Politècnica de Cataluña
2001 – 2007
Thesis: Multistage scenario optimization procedure to plan annualized working hours with uncertainty factors.

COURSES

- Deep Learning Specialization
- Building Essential Leadership Skills
- Clear Leadership
- Interpersonal Communication and Leadership

LANGUAGES

- English: native
- Spanish: native
- Swedish: fluent